

◆ Gemini

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow # Import cv2_imshow

# Frame properties
width, height = 640, 480
fps = 30
duration = 30 # seconds
total_frames = fps * duration

# Ball position and speed
x, y = 100, 100
dx, dy = 5, 4

for i in range(total_frames):

    # Create black frame
    frame = np.zeros((height, width, 3), dtype=np.uint8)

    # Draw moving ball
    cv2.circle(frame, (x, y), 30, (0, 255, 255), -1)

    # Draw moving rectangle
    cv2.rectangle(frame, (200, 150), (350, 280), (255, 0, 0), 3)

    # Draw moving line
    cv2.line(frame, (0, i % height), (width, (i + 150) % height), (0, 255, 0), 2)

    # Display frame number
    cv2.putText(frame, f"Frame: {i+1}", (20, 40),
                cv2.FONT_HERSHEY_SIMPLEX, 1, (255,255,255), 2)

    # Update ball position
    x += dx
    y += dy

    if x >= width-30 or x <= 30:
        dx = -dx
    if y >= height-30 or y <= 30:
        dy = -dy

    # Source points
    src = np.float32([
        [100,100],
        [540,100],
        [100,380],
        [540,380]
    ])

    # Destination points
    dst = np.float32([
        [50,50],
        [590,100],
        [120,430],
        [520,380]
    ])

    # Perspective Transformation
    M = cv2.getPerspectiveTransform(src, dst)
    output = cv2.warpPerspective(frame, M, (width, height))
    cv2_imshow(frame)
    cv2_imshow(output)
```


Frame: 1



Frame: 2















